# Correlation Test Among EMI Measurement Sites Using a Comb Generator

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**Abstract**

VCCI had considered that a major reason why we cannot get correlated data on the radiated interference (especially one from the power cord in the frequency range up to 300 MHz) among different measurement sites might be that the characteristics and environment of the power lines to the EUT (equipment under test) are different at each site. Therefore, we performed a correlation test at eight EMI measurement sites using a comb generator as an EUT to whose output terminal a cable was connected, based on an assumption that the correlation among sites would be improved if the common-mode impedance of the power lines is specified at each site. The test results show that the correlation among sites was improved when the common-mode impedance of the power lines was constant.

But, the actual laying route and position of the power cord had some influence in the frequency range of 60 to 80 MHz, and the effect of the constant common-mode impedance was less clear than one in the other frequency ranges.

Also, the correlation among 14 sites using an actual EUT was improved comparing the standard deviations with the current CISPR22 measurement conditions, when the common-mode impedance of the power lines to an EUT was constant.

## 1.Tests with a comb generator

In this test, we paid attention only to the interference radiated from the power cord. So, to simulate an EUT that radiates an interference from the power cord, we adopted a comb generator to whose output terminal a two-parallel-wire cable was connected through a balun. With this comb generator, we made measurements in seven semianechoic chambers and one open-area test site.

The measurement layout and conditions were as follows (refer to Fig. 1-3):

(1) Layout and conditions of the power cord
- Distance between the balun output and the table edge: 75 cm
- Vertical distance between the table edge and a ground plane: 70 cm
- Distance between the first cord contact on the ground plane and the cord terminating point: 75 cm

(2) Three different conditions at the cord terminating point
a) The end of the power cord was directly connected to the power lines to the EUT of each site. The EUT power source was kept off (Fig. 1).
b) Each of the ends of the two-line power cord was terminated with 50Ω resistance (Fig. 2).
c) Each of the ends of the two-line power cord was kept open (Fig. 3).

For conditions b) and c), a terminating tool was used. It was placed at the center of the turntable.

The measurement conditions were as follows:

(1) Measurements were taken at a distance of 3m.

(2) Measurements were taken at ten measurement frequencies: 40, 50, 60, 70, 80, 100, 120, 150, 200, and 270 MHz

(3) Antenna height was varied from 1 to 4m. The turntable was rotated to angles of 0 to 360 degrees with respect to the antenna. The maximum electrical field strength was measured as a quasi-peak value at both horizontal and vertical polarization.

The maximum electrical field strength of the horizontal and vertical polarization were measured at each test site and each frequency.

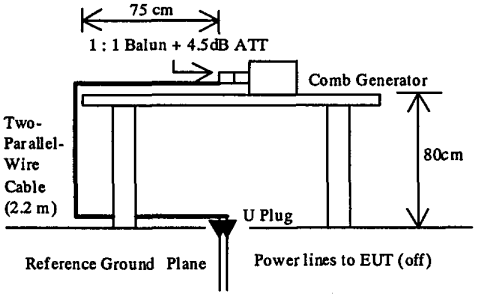



**Fig. 1 Measurement layout: EUT directly connected to power lines**

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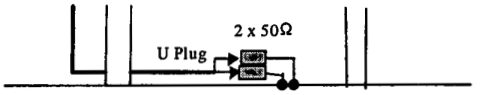



**Fig. 2 Measurement layout: EUT terminated with 50Ω**

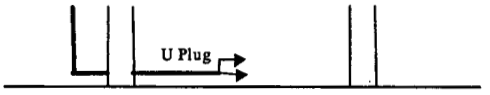



**Fig. 3 Measurement layout EUT with open circuit**

## 2.Results

The standard deviation of the maximum electrical field strength for each terminating condition is graphed in Fig. 4. The average value of the maximum electrical field strength for each terminating condition is graphed in Fig. 5.

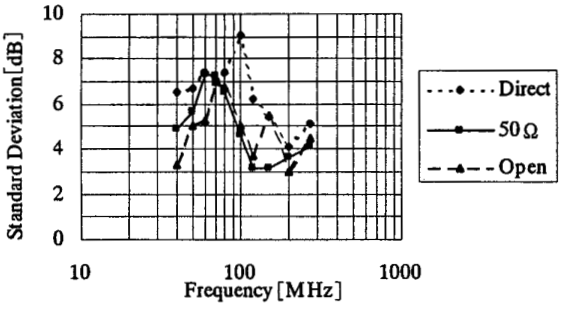



**Fig. 4 Standard deviation among measurement sites for each terminating condition**

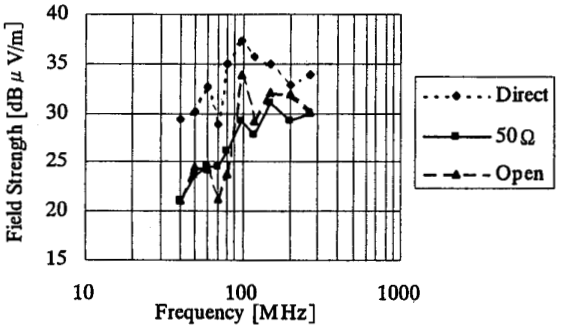



**Fig. 5 Average of maximum field strength at the eight sites for each terminating condition**

As Fig. 4 shows, the correlation among sites improved when the impedance of the power lines to the EUT of each site was constant. At frequencies from 60 to 80 MHz, however, there was no big difference between the standard deviation for constant power line impedance and that for a power cord directly connected to the power lines to the EUT. This result makes it clear that, at these frequencies area, the differences in the characteristics and physical position of the power lines outlet to the EUT has an influence to the correlation of the radiated interference measurements mainly from the power cord. To investigate the standard deviation showed so little difference at frequencies from 60 to 80 MHz, we performed an additional experiment, described in the next section.

Fig. 5 also shows that the electrical field strength measured for the power cord directly connected to the power lines to the EUT was different from that measured when the power line impedance was constant. For the 50Ω termination, the comb generator output with 50 Ω impedance was connected to a power cord via a balun and 50Ω terminating resistors. That condition made better matching from the output of the comb generator to the power lines than when the power cord was directly connected to the power lines to the EUT. And in this condition, the electrical field strength seemed lower than with the direct connection, over the entire range of measurement frequencies. For the open condition also, the electrical field strength did not differ much from that for the 50Ω termination condition.

The common-mode impedance of the power lines at each measurement site might vary from several ohms to several hundred ohms. If there were a strong correlation between the power line impedance and the electrical field strength of the radiated interference, one would expect the electrical field strength of the radiated interference to differ along with the measurement conditions. Our results, however, showed no such correlation.

There is a need for further study to determine how the common-mode impedance of the power lines is related to the electrical field strength of the radiated interference.

## 3.An additional experiment

In an effort to determine why we could not get improved correlation among sites at frequencies of 60 to 80 MHz, we made the following additional test at one of the sites where the correlation tests were performed:

(1) A test to investigate the influence of different positions of the outlet of the power lines to the EUT.

### *3.1 Positions of the outlet of the power lines to the EUT*

This experiment was performed to verify how the position of the outlet influenced the electrical field strength when the power cord was directly connected to the power lines to the EUT. The positions of the outlet were arranged to simulate the actual positions of the outlet at each site.

The conditions of the measurement were as follows:

(1) The termination was done with 50Ω.

(2) The measurements were made at the same 10 frequencies as in the correlation test.

(3) Vertical polarization was measured, because in the correlation test the maximum electrical field strength

had usually been observed in the vertical polarization.

(4) The antenna height was changed from 1 to 4 m, the turntable was rotated to angles of 0 to 360° to the antenna, and the maximum electrical field strength was measured as the quasi-peak value.

(5) The positions of the 50Ω termination are shown in Fig. 6. (The position of the power cord was the same for all three cases, up to the point where it contacted the ground plane.)

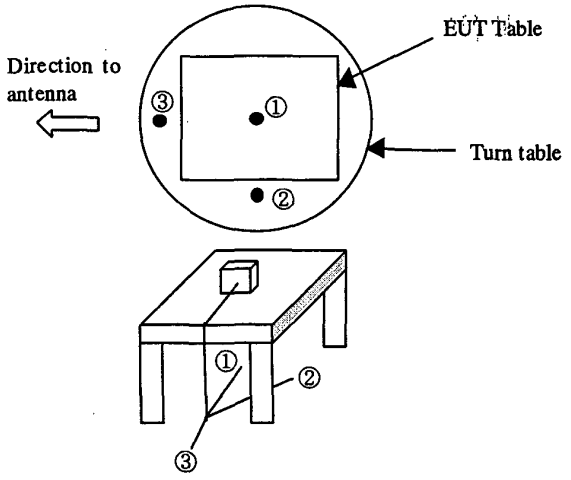



**Fig.6 Layout of the experiment: influence of outlet positions**

### *3.2 Results of the experiment on the influence of outlet positions*

The deviation of the electrical field strength (the maximum strength minus the minimum strength) for the different outlet positions is shown in Fig. 7. Clearly, outlet position has a big influence, especially at 50, 60, and 80 MHz. This indicates that not only the power line impedance, but also the position of the power line outlet, influenced the data when the power cord was directly connected to the power lines to the EUT in the site correlation tests. From this result we can surmise one reason why the radiation pattern of the electric field varies. The coupling of the power cord to the ground plane forms a geometric shape, and so does the position of the power cord on the ground plane. As the turntable is rotated, these shapes change, and so the radiation pattern of the electric field from them changes too.

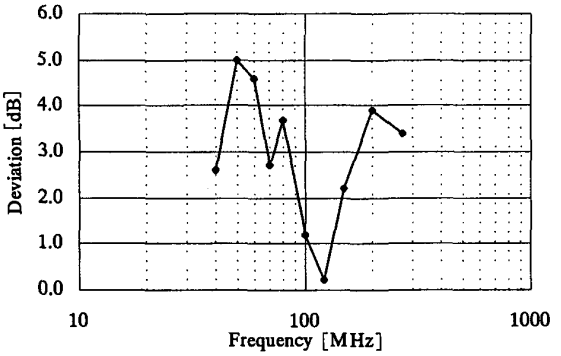



**Fig. 7 Influence of the power line outlet positions**

### 4. A site correlation test using an actual EUT

We performed a site correlation test using an actual machine, based on the results of the tests described above. In this test, to make the radiation from the power cord dominant, we adopted as an EUT a keyboard that incorporated a CPU and that had no display connected. That is, only a power cord was connected to the EUT. We adopted two conditions for the power cord connection. One was the condition specified in CISPR22. The other was that the EUT was connected to the power lines through a Line Impedance Stabilization Network (LISN) as shown in Fig. 8 [1]. The LISN had a common-mode constant impedance of 50Ω up to around 500 MHz on the power line side. We tested the correlation with each of the connecting conditions described earlier at 14 test sites. For this measurement, we selected 12 frequencies that had a dominant level of narrow band interference radiated from the EUT from 30 to 150 MHz. To reduce the influence of the positions of power lines to the EUT as much as possible, we positioned the tool at the center of the turntable and supplied the power to the EUT at that position for the condition of constant common mode impedance at each site.

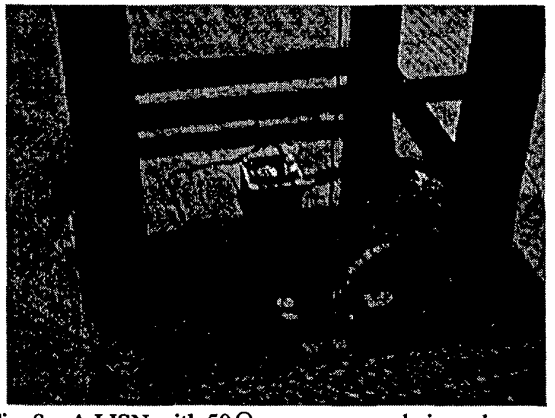


Fig. 8 A LISN with 50Ω common-mode impedance on power line side

### *4.1 Results*

The radiated interference measured at 14 sites with two conditions is graphed in Fig. 9 and Fig. 10.

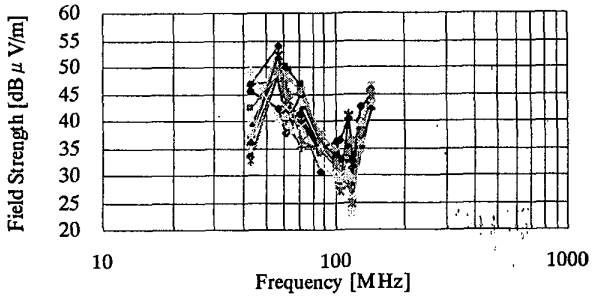



**Fig. 9 Measured data at 14 sites: for the connection as specified in CISPR22**

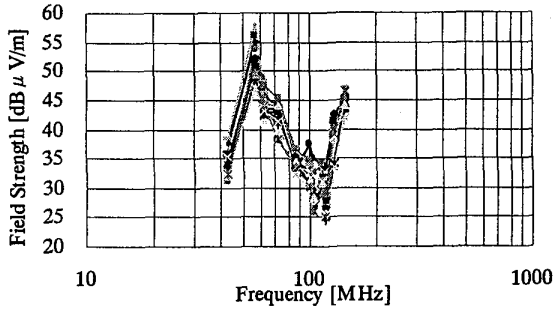



**Fig. 10 Measured data at 14 sites: for connection to 50Ω common-mode impedance**

The deviation among sites is shown in Fig. 11, and the average electrical field strength for the 14 sites is shown in Fig. 12.

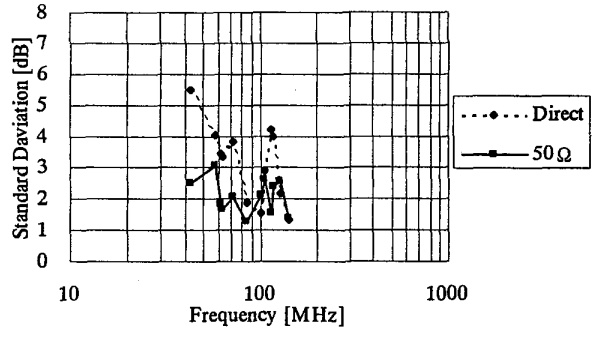



**Fig.11 Standard Deviation among sites of electrical field strength**

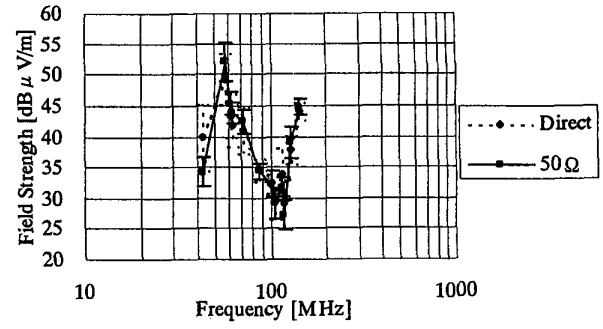



**Fig.12 Average of electrical field strength for the 14 site**

The above results show that the standard deviation for the 14 sites was almost within 3 dB for radiated interference under 100 MHz when common-mode impedance was specified to a constant of 50Ω (Fig. 10). This standard deviation was a great improvement over the one obtained by the current CISPR22 measurement method. For the average electrical field strength, Fig. 12 shows almost no difference between the two conditions. This result proves that making the common-mode impedance of the power lines to the EUT constant was one important factor in improving correlation among sites.

## Conclusions

In correlation tests with a comb generator and an actual EUT, we found that specifying the impedance of the power lines to the EUT improves the correlation among measurements of radiated interference taken at different sites. We also found that the positions of the power lines to the EUT were different among sites with the current measurement method and this difference was a factor in degrading the correlation among sites. This may require new wiring specifications on the turntable, to reduce the influence of the power cord and the reference ground plane.

Currently there is still a deviation about 10 dB maximum among sites even when we eliminate the above factors. In future, it is important to improve the characteristics of measurement facilities such as semianechoic chambers, and to reduce uncertain factors including antenna-factors, cable losses, and receivers for the electrical field measurement required for the radiated interference measurement. There should also be a verification to make the relation between the power line impedance and the electrical field strength of the radiated interference clear.

## Reference

[1] Maeda.A:"Stabilization of Power Line Impedance for Radiated EMI Level Measurement", IECIE Trans.Commun. Vol.E75-B, No.3 March 1992 , pp.148-156